

| Option/Variable              | Meaning           | Function                                                                                                               |
|------------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>Options</b>               |                   |                                                                                                                        |
| -e <date>                    | Expire            | Expiration date in dd/mm/yyyy format [none]                                                                            |
| -m <message>                 | Message           | Message to display upon expiration ['Please contact your provider']                                                    |
| -f <script>                  | File              | File name of the script to compile                                                                                     |
| -i <shell options>           | Inline            | Inline option for the shell interpreter, i.e., -e                                                                      |
| -x <command>                 | Exec              | Exec command, as a printf format, i.e., exec(\'%s\'.@ARGV);                                                            |
| -l <Option>                  | Last              | Last shell option, i.e., --                                                                                            |
| -r                           | Relax             | Relaxes security. Makes a redistributable binary which executes on different systems running the same operating system |
| -v                           | Verbose           | Verbose compilation                                                                                                    |
| -D                           | Debug             | Switches on debug exec calls [OFF]                                                                                     |
| -T                           | Traceable         | Allows binary to be traceable (using strace, ptrace, truss, etc)                                                       |
| -h                           | Help              | Displays help and exits                                                                                                |
| -C                           | Copyright licence | Displays licence and exits                                                                                             |
| -A                           | Abstract          | Displays abstract and exits                                                                                            |
| <b>Environment variables</b> |                   |                                                                                                                        |
| \$CC                         | C compiler        | C compiler command [cc]                                                                                                |
| \$CFLAGS                     | C Flags           | C compiler flags [none]                                                                                                |

Table 1: SHC options and variables

available to execute the script.

#### 4. Setting an expiry date to the encrypted shell script

Using SHC, you can also specify an expiry date. When somebody tries to execute the shell script after this date, they'll get an error message.

If you don't want anybody to execute the `welcome.sh.x` after December 31, 2014 (I used an earlier year's date for testing purposes), create a new encrypted shell script using the `shc -e` option to specify the expiry date, which needs to be specified in the dd/mm/yyyy format.

```
[root@test src]# shc -e 31/12/2014 -f welcome.sh
```

In this example, if someone tries to execute `welcome.sh.x` after December 31, 2014, they'll get a default expiry message as shown below:

```
[root@test src]# ./welcome.sh.x
./welcome.sh.x: has expired!
Please contact your provider
```

To display custom expiration messages, use the `-m` option (along with the `-e` option as shown below):

```
[root@test src]# shc -e 31/12/2014 -m "Contact suresh@
linuxsecret.com for new version of this script" -f welcome.sh
[root@test src]# ./welcome.sh.x
./welcome.sh.x: has expired!
```

Contact [suresh@linuxsecret.com](mailto:suresh@linuxsecret.com) for new version of this script.

#### 5. Creating redistributable encrypted shell scripts

You can create redistributable encrypted shell scripts using the `-r` (relax security) and `-T` (traceable using `strace`) options:

```
[root@test src]# shc -v -r -T -f welcome.sh
shc sh1=sh
shc [-i]=-c
shc [-x]=exec '%s' "$@"
shc [-l]=
shc opts=
shc: cc welcome.sh.x.c -o welcome.sh.x
shc: strip welcome.sh.x
shc: chmod go-r welcome.sh.x
```



#### References

- 1] <http://www.datsi.fi.upm.es/~frosal/>
- 2] [https://en.wikipedia.org/wiki/Shc\\_the\\_shell\\_script\\_compiler](https://en.wikipedia.org/wiki/Shc_the_shell_script_compiler)
- 3] <http://tech.michaelatfield.net/2009/06/13/install-build-essential-on-rhelcentos/>
- 4] <http://linux.die.net/man/1/shc>

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